

 Announcement

Kindly note: We're currently experiencing a high volume of inquiries and calls, potentially leading to delays in our initial response.



Dermatology 23S Module 1

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Day 3 task

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by [Huma Jaffar \(Tutor\)](#) - Sunday, 8 October 2023, 6:02 AM

Day 3 task

Q 1- There have been notorious cases in the press of GPs facing manslaughter charges for missing the diagnosis of Addison's disease - see attached article!

<https://www.pulsetoday.co.uk/news/regulation/two-gps-accused-of-manslaughter-after-death-of-child-with-addisons-disease/>

- What are the dermatological features of Addison's disease?
- Is there a useful screening test that could be employed in primary care?
- Why is hyperpigmentation a feature of Addison's?

Q 2- A 40-year-old man had been referred over with a new diagnosis of diabetes, having presented to his GP with polyuria and weight gain. His GP had checked his fasting glucose and found this to be 18.1 mmol/l (3.5-5.5). He provided no other history of note, other than his mother having had type 2 diabetes diagnosed at the age of 50. He had gained weight over the last 5 years and had a BMI of 35 kg/m². There was nothing otherwise of note on examination.

However, as he was being examined, he mentioned that he had been aware of a rash on his buttocks consisting of raised, non-tender papules as shown. He had mentioned this to his GP, who had explained to him that these were probably like a 'nappy rash' and to use 'cleansers and emollients'. The lesions persisted. These had been present for about 1 year.



- If you could request one single investigation, what would it be? List your differential diagnoses.

222 words

Reply



Re: Day 3 task

by [Pooran Outar](#) - Monday, 9 October 2023, 5:12 AM

Case 1: Addison's Disease

Question 1 - Dermatological features of Addison's disease

- Hyperpigmented scar on diffusely hyperpigmented skin frequently on knuckles or the skin creases nipples, axillae, perineum, and buccal mucosa (Liotta, 2022)



(Image 1)

- Oral mucous membrane hyperpigmentation is pathognomonic for the disease (Liotta, 2022).



(Image 2)

- Vitiligo
- Decreased pubic and axillary hair in women

- Longitudinal melanonychia

Question 2 - Useful diagnosis in Primary Health Care

In PHC, diagnosis can be guided by a **Basic Metabolic Panel** and the disease's clinical presentation. Common findings include hyponatremia (90% of patients), hyperkalemia (60-70% of patients), and rare hypercalcemia (5-10% of patients). definitive diagnosis is achieved by (Liotta, 2022).

I choose this test over the **insulin tolerance test**, which is sensitive for adrenal insufficiency because it requires close monitoring of the patient and maybe contraindicated in some cases.

I am interested in knowing if there is an established screening test for this disease, as I am unaware of such.

Question 3 - Why is hyperpigmentation a feature of Addison's?

"Hyperpigmentation is the physical finding most characteristic of Addison disease, arising from continual stimulation of the corticotrophs in the anterior pituitary. Specifically, it results from cross-reactivity between the ACTH produced by the corticotrophs and the melanocortin 1 receptor on keratinocytes.

Hyperpigmentation is usually generalized over the entire body and can be found in palmar creases, buccal mucosa, vermilion border of the lips, and around scars and nipples. It is not a feature of secondary adrenal insufficiency because of the lack of increased ACTH in these patients" (Michels and Michels, 2014).

Reference list:

- Liotta, E.A. (2022). *Dermatologic Aspects of Addison Disease Workup: Laboratory Studies, Imaging Studies, Histologic Findings*. [online] emedicine.medscape.com. Available at: <https://emedicine.medscape.com/article/1096911-workup>.
- Michels, A. and Michels, N. (2014). Addison Disease: Early Detection and Treatment Principles. *American Family Physician*, [online] 89(7), pp.563-568. Available at: <https://www.aafp.org/pubs/afp/issues/2014/0401/p563.html>.

302 words

Reply



Re: Day 3 task

by [Adam Tollitt](#) - Monday, 9 October 2023, 9:06 AM

Hi Pooran,

Following on your answer about screening tests for Addison's Disease, in primary care if there is clinical suspicion of Addison's disease in a stable patient then typically an early morning serum cortisol level is taken (ideally between 8am-9am to capture the natural serum cortisol peak).

Exact interpretation of results depends on local laboratory reference ranges but typically if the early morning serum cortisol level is >500 nmol/L then primary adrenal insufficiency can be excluded (NICE CKS, 2020).

Confirmatory testing of Addison's disease is undertaken in secondary care typically with a Synacthen test. Synthetic ACTH is given to stimulate the adrenal gland and serial measurements of serum cortisol taken to assess for an appropriate rise in serum cortisol (or in the case of Addison's an inadequate serum cortisol rise).

References

National Institute for Health and Care Excellence. (2020). Addison's disease – How should I investigate suspected adrenal insufficiency in Primary Care. [online]. <https://cks.nice.org.uk/topics/addisons-disease/diagnosis/investigations-suspected-adrenal-insufficiency/> (accessed 09/10/2023)

155 words

Reply

**Re: Day 3 task**by [Sadaf Basharat](#) - Wednesday, 11 October 2023, 11:11 PM

The most common tests for Addison's are well discussed by my colleagues. I just wanted to add in the discussion another test to look for Addison's.

The **Metyrapone** test.

The Metyrapone test works by interrupting the cortisol production process through the inhibition of the enzyme 11 β -hydroxylase, responsible for converting 11-deoxycortisol (11-s) into cortisol. At midnight, a dose of Metyrapone (30 mg/kg) is administered intravenously, and then, cortisol and 11-s levels are measured 8 hours later. A normal outcome is characterized by an elevation in serum 11-s levels exceeding 7 mg/dL. When 11-s levels remain below this threshold, it can suggest the presence of adrenal insufficiency.

Reference:

Elizabeth A Liotta (2022), Dermatologic Aspects of Addison Disease Workup, Medscape Available at <https://emedicine.medscape.com/article/1096911-workup#?form=fpf>

121 words

Reply

**Re: Day 3 task**by [Xiao Liey Ding](#) - Monday, 9 October 2023, 4:14 PM

Hi,

Serum 8 am cortisol is a useful screening test for patient suspected to have adrenal insufficiency in primary care setting. If the morning cortisol level $> 370\text{nmol/L}$, a diagnosis of adrenal insufficiency is made unlikely. If the 8 am cortisol level is low or there is still strong clinical suspicion despite normal result, Short Synacthen Test can be performed to confirm the diagnosis. Blood will first be taken at 0-hour for cortisol and ACTH. Synacthen 250mcg then given intravenously (alternative IM) to the patient and again blood taken at 0.5 hr and 1 hr for cortisol level. Patient need to stop taking steroid- containing medication for 24-48 hr prior to SST in order to produce an accurate test result. Generally, a cortisol level < 550 at 0.5 hr is considered inadequate (I can't find conclusive answer from papers. Guess it depends largely on laboratory reference range. This level is based on my vague recollection from my previous Int Med posting).

Reference:

Wee Shuan Yo, Li-Mae Toh, Suzanne J. Brown, et al. How good is a morning cortisol in predicting an adequate response to intramuscular synacthen stimulation? Clinical Endocrinology. July 2014. Volume 81, Issue 1, pg 19-24. Available at: <https://doi.org/10.1111/cen.12373>

200 words

Reply

**Re: Day 3 task**by [Siddharth Saluja](#) - Wednesday, 11 October 2023, 12:31 PM

Hello Xiao,

Thanks for the above post. I just wanted to get your thoughts from your experience, classically we all recognise Addisons as hyponatraemia, hyperkalemia and hyperpigmentation, when would it be appropriate to refer to secondary care if patients presents with no clinical signs presents in primary care but has those findings in their bloods tests? I ask this because in primary care we often get blood disturbances without any clinical symptoms and often when we call them for a telephone review or call them into the surgery, they are often confused as to why they were called in when they have no symptoms of adrenal insufficiency. I suppose as primary care physicians we have to be overall cautious and check for adrenal sufficiency and start them on the right treatment replacement of hydrocortisone/fludocortisone if needed but there isn't always a clear guidance on this. Was trying to get you thought on your experiences on secondary care referrals made for asymptomatic patients made on blood findings?

168 words

Reply



Re: Day 3 task

by [Xiao Liey Ding](#) - Wednesday, 11 October 2023, 5:44 PM

Hi Siddharth,

I have not encountered patients with suspected Addison disease in primary care so far. Hence, below is just my 20 cents based on reading and understanding and a few cases where I had the chance to participate during my time in secondary care.

Addison disease has slow onset nature and its non-specific early symptoms such as fatigue and muscle weakness can make detection very hard and goes undiagnosed for long time. Over time, the symptoms may become more severe and patient may experience further symptoms. *Recurrent* hyponatremia and hyperkalemia in repeat blood test after cessation of correction treatment should prompt suspicion of diagnosis and ought to be further investigated with 8am cortisol and ACTH (Short Synacthen Test in borderline 8am cortisol or ACTH case). If the diagnosis of adrenal insufficiency is confirmed with blood tests, the underlying cause need to be ascertained and referral to endocrinologist is indicated.

MSD manual stated majority of cases are due to autoimmune disease with remaining result from adrenal destruction secondary to tumor, granulomatous infection (histoplasmosis/ tuberculosis) or hemorrhage (Ashley B. Grossman, 2022). Hence, referral to endocrinology colleague after confirmation of diagnosis is essential despite patient has no symptoms. The aim is to identify and treat the underlying etiology and prevent serious complication such as adrenal crisis with mineralocorticoids and glucocorticoid replacement. In children, congenital adrenal hyperplasia is the commonest cause for Addison disease.

The final diagnosis is often made when we review the case retrospectively, which sometimes after we found adrenal incidentaloma on CTAP together with blood test abnormality, or patient is acutely sick in adrenal crisis.

Reference:

Ashley B. Grossman (2022). Available at: <https://www.msdmanuals.com/professional/endocrine-and-metabolic-disorders/adrenal-disorders/addison-disease#v982347> (Accessed: 11 Oct 2023)

278 words

Reply



Re: Day 3 task

by [Pooran Outar](#) - Monday, 9 October 2023, 5:34 AM

Case 2

My presumptive diagnosis is Eruptive Xanthomatosis

My investigation of choice is Lipid profile.

Differential diagnosis (Emanuel and Palma, 2048):

- Leprosy infection histological analysis may reveal an accumulation of foamy histiocytes in the dermis and special stains can identify the causative micro organisms
- Other xanthomas which normally do not have a high level of extracellular lipid and an acute inflammatory response.
- Xanthogranulomas which are characterized histologically by Touton giant cells

In a case study similar to this Pierce, Patel and Scott, 2021 also suggest the following differential diagnosis:

- Multicentric reticulohistiocytosis
- Disseminated granuloma annulare
- Sarcoidosis

Reference list:

- Emanuel, P. and Palma, R. (2048). *Eruptive xanthoma pathology* / *DermNet*. [online] dermnetnz.org. Available at: <https://dermnetnz.org/topics/eruptive-xanthoma-pathology> [Accessed 9 Oct. 2023].
- Pierce, J., Patel, T. and Scott, C. (2021). Eruptive Xanthomas. *Mayo Clinic Proceedings*, [online] 96(12), pp.3097–3098. doi:<https://doi.org/10.1016/j.mayocp.2021.09.019>.

135 words

Reply



Re: Day 3 task

by [Zarmeena Khan](#) - Wednesday, 11 October 2023, 3:03 PM

Q2.I would consider diagnosis of Eruptive xanthomatosis ; a skin condition often associated with poorly controlled diabetes mellitus, particularly type 2 diabetes. It is characterized by the development of small, yellow, firm, and often itchy bumps or nodules on the skin. These skin lesions are a result of the deposition of lipids, including triglycerides and cholesterol, in the skin.

This occurs when high levels of triglycerides in the blood lead to the accumulation of lipids in the skin. The exact mechanism is not fully understood, but it is believed to be related to insulin resistance, impaired lipid metabolism, and increased serum triglyceride levels in individuals with poorly controlled diabetes.

Patients with eruptive xanthomatosis typically present with crops of small, yellow or flesh-colored papules or nodules that may be surrounded by erythema. These lesions are often pruritic (itchy) and can appear on the buttocks, elbows, knees, hands, feet, and other areas. They can range in size from a few millimeters to a few centimeters.

b. Testing:

Diagnosis is often based on the clinical appearance of skin lesions and is supported by laboratory tests that reveal elevated serum triglyceride levels. It's essential to differentiate eruptive xanthomatosis from other skin conditions, such as xanthelasma, which occurs around the eyes and is not associated with hypertriglyceridemia.

c. Differentials :

While it is often associated with poorly controlled diabetes mellitus and hypertriglyceridemia, several other skin conditions and systemic diseases may present with similar skin lesions. The differentials of eruptive xanthomatosis include:

1. **Xanthelasma:** Xanthelasma is a condition in which yellowish plaques or nodules form around the eyes. These are due to lipid deposits and can also be associated with hyperlipidemia but are distinct from eruptive xanthomatosis.
2. **Xanthoma Disseminatum:** This rare non-Langerhans cell histiocytosis can lead to the development of multiple, reddish-brown papules on the skin. It is often associated with diabetes insipidus and hyperlipidemia.
3. **Eruptive Keratoacanthomas:** While less common, these are skin lesions that can resemble eruptive xanthomatosis. Eruptive keratoacanthomas are fast-growing, dome-shaped nodules that typically appear on sun-exposed areas.
4. **Cutaneous Lichen Planus:** Lichen planus can cause raised, itchy, reddish or purplish bumps on the skin, which may be mistaken for eruptive xanthomatosis.
5. **Nodular Prurigo:** This is a skin disorder characterized by the development of itchy, dome-shaped nodules on the skin, which can sometimes be confused with eruptive xanthomatosis due to their pruritic nature.
6. **Prurigo Pigmentosa:** This is a rare inflammatory skin condition that can lead to the formation of pruritic, erythematous papules, and plaques. In some cases, it may appear similar to eruptive xanthomatosis.
7. **Cholesterol Emboli:** In rare cases, cholesterol emboli can lead to skin lesions that may be confused with eruptive xanthomatosis. These emboli can be a result of vascular disease.

The primary goal in the management of eruptive xanthomatosis is to address the underlying metabolic abnormality, which is often uncontrolled diabetes and hypertriglyceridemia. Management strategies may include improving blood glucose control and lipid-lowering therapy (e.g., statins, fibrates) .

References:

Greenberg AS, McDaniel ML. (2001). *Identifying the links between obesity, insulin resistance and beta-cell function*: potential role of adipocyte-derived cytokines in the pathogenesis of type 2 diabetes. *Eur J Clin Invest*, 31(10), 972-979.

Hober D, Soria A, Guerrero D. (2019). *Xanthomas: Eruptive Xanthomas in Diabetes Mellitus*. *Pathobiology*, 86(5-6), 308-316.

Skov MW, Gniadecki R. (2014). *Eruptive xanthomas in diabetes*. *J Diabetes Investig*, 5(6), 749-750.

548 words

Reply



Re: Day 3 task

by [Jillian Simpson](#) - Wednesday, 11 October 2023, 4:14 PM

My initial impression is that this is an eruptive xanthomas secondary to hypertriglycerdaemia.

My one single investigation would be a fasting lipids profile.

Poorly controlled diabetes can significantly increase triglyceride levels, which can result in eruptive xanthomas. Eruptive xanthomatosis is a typical but rare sign of acute hypertriglyceridemia. It presents as a sudden appearance of yellowish firm papules located on the buttocks and extensor areas of extremities (Martinez et al, 2008).

Another differential would be xanthoma dissemination. I have included a picture of this from a journal article from Behera et al (2023). This tends to occur with mucosal xanthoma too.

Other differentials to consider include tuberous and tendon xanthomas, verruciform xanthoma and plane xanthomas (Ngan, 2020).

Martínez, D., Díaz, J.Ó. and Bobes, C. (2008). Eruptive xanthomas and acute pancreatitis in a patient with hypertriglyceridemia. *International Archives of Medicine*, 1(1), p.6. doi:<https://doi.org/10.1186/1755-7682-1-6>.

Wanat, K., Noe, M. (2022). *UpToDate*. [Online] Available at: https://www.uptodate.com/contents/cutaneous-xanthomas?search=eruptive%20xanthomas&source=search_result&selectedTitle=1~150&usage_type=default&display_ [Accessed 11 Oct. 2023].

Behera, B., Malathi, M., Thappa, D.-M., Vamanshankar, H., Parida, P.-K. and Gochhait, D. (2017). Xanthoma Disseminatum Presenting with Hoarseness. *Iranian Journal of Otorhinolaryngology*, [online] 29(95), pp.365–368. Available at: <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC5785117/> [Accessed 11 Oct. 2023].

Ngan, V. (2020). *Xanthoma* / *DermNet NZ*. [online] Available at: <https://dermnetnz.org/topics/xanthoma>. [Accessed 11 Oct. 2023].

205 words



Reply



Re: Day 3 task

by [Sophie Holloran](#) - Thursday, 12 October 2023, 9:53 PM

Hi Jill and Zarmeena,

I agree with your differential, although I must say I wouldn't have been able to suggest this myself without reading first - I've not come across this presentation myself.

I wonder if there is a threshold of poor HbA1c control that makes this more likely? And how long does someone have to have bad control for before they develop these lesions?

I really like your differential list Zarmeena. I wonder as well, if you would consider neurofibromatosis on this list? I think it would be quite unusual to have localised lesions in NF but I think some of the neurofibromas could look quite similar?

108 words

Reply



Re: Day 3 task

by [Jillian Simpson](#) - Friday, 13 October 2023, 8:19 PM

Hi Sophie,

I must admit I did the reading from the reference list before attempting this week's cases. I have seen some very high triglycerides readings in clinical practice. One of which was a patient with uncontrolled diabetes drinking a lot of alcohol so difficult to say if it was a combination of the two or the length of time his diabetes was uncontrolled. I don't remember there being any skin manifestations with it though.

The consultant I work with also remembers a case during her training when a patient died from pancreatitis secondary to high triglycerides. This patient was taking a retinoid which has made me very cautious with monitoring those on isotretinoin in our acne clinics.

118 words

Reply



Re: Day 3 task

by [Bhavika Patel](#) - Saturday, 14 October 2023, 4:32 PM

Hi Jillian,

I was just thinking the same. I too have an increasing number of patients with raised triglyceride levels especially ones who refuse to start lipid-reducing therapy. I haven't noticed any evidence of eruptive xanthomas but will admit to not going out my way to look for them in a busy clinic. I have also noticed patients with xanthlasma but with a normal lipid profile despite no lipid-reducing therapy which has been odd.

I found that eruptive xanthomas occur with triglyceride levels of >1000 mg/dL which may be the reason and this occurs in roughly 18/100,000 cases (Hsueh, YC. 2019)

Hsueh, Y-C. Chou, C-L. Lee, T-I. (2019). Diabetic dyslipidemia with eruptive xanthoma. Cleveland Clinic Journal of Medicine [online]. 86 (9) 575-576; DOI: 10.3949/ccjm.86a.18104 (Accessed on 14th October 2023)

129 words

Reply



Re: Day 3 task

by [Arwaf Benrawwaf](#) - Sunday, 15 October 2023, 8:04 PM

Thank you Sophie, for your question regarding the inclusion of neurofibromatosis on the list of potential diagnoses.

Neurofibromatosis (NF) is a genetic disorder that primarily affects the nervous system, causing tumors to form on nerve tissue. There are three types of neurofibromatosis: NF1, NF2, and schwannomatosis.

In regards to your question about whether neurofibromatosis should be considered for a list of conditions with localized lesions, it is important to note that neurofibromas can indeed present as localized lesions. Neurofibromas are benign tumors that develop from the cells that surround nerves. They can occur anywhere in the body and vary in size and appearance.

In NF1, which is the most common type of neurofibromatosis, individuals may develop multiple neurofibromas on or under the skin. These tumors can appear as soft, fleshy growths or bumps and can be similar in appearance to other localized lesions.

However, it is essential to differentiate between neurofibromas associated with NF1 and other types of localized lesions. Diagnosis of NF1 typically involves family history, clinical examination, and genetic testing,

References:

Friedman, J.M. (2002). Review Article : Neurofibromatosis 1: Clinical Manifestations and Diagnostic Criteria. *Journal of Child Neurology*, 17(8), pp.548-554.

doi:<https://doi.org/10.1177/088307380201700802>.

194 words

Reply

**Re: Day 3 task**by [Huma Jaffar \(Tutor\)](#) - Thursday, 12 October 2023, 9:54 PM

I like the pictorial posts! welldone group

For addisons - as a primary care provider - in addition to clinical signs, a good screening test would be an 8 am cortisol level!

This patient with poorly controlled diabetes has eruptive xanthomatosis- Great pick up of diagnosis !
Agreed that the diagnosis is confirmed by requesting lipids and one would expect a turbid specimen with very high triglycerides.
for ddx Molluscum contagiosum or indeed dermatitis herpetiformis may be on that list. A shave biopsy could confirm the diagnosis with extracellular lipid deposition. It is important to make this diagnosis as there is a particularly high risk of acute pancreatitis.

108 words

[Reply](#)

**Re: Day 3 task**by [Mostafa Damghani](#) - Sunday, 15 October 2023, 7:38 PM

1- Dermatological features of addison disease are a generalised darkening of the skin or hyperpigmentation. the skin in general becomes significantly darker and that can affect the tongue as well. this is mainly due to the negative feed back due to cortisol deficiency to the hypothalamic pituitary adrenal axis, which then leads to increased levels of prohormone called proopiomelanocortin which then is cleaved to melanocyte-stimulating hormone increasing melanin production and causing hyperpigmentation all across the body.

2- addison disease can be an autoimmune disease, taking extra caution in patients that have existing autoimmune disorders such as thyroid, vitiligo coeliac diseases and etc. we can follow ADDISONS mnemonic to look for signs of addisons disease in primary care

A: Always tired, Abdominal pain, Ache in Limbs and back

D: Dizzy on standing

D: Drop in blood pressure when standing

I: Inexplicable weight loss

S: Skin color change, specially where there is more friction, such as knuckles , elbows and sun-exposed areas.

O: Only eating sparingly (anorexia)

N: No strength in Hands and limbs

S: Sick or nausea

looking for biochemical abnormalities such as hyponataemia and hyperkalaemia, hypercalcaemia and hypoglycaemia

unexplained hyponatraemia should flag adrenal insufficiency

Haematological abnormalities such as mild normocytic anaemia should also flag further check for addisons disease

and testing for the levels of cortisol (9am) if less than 100nmol/L can possibly indicate addison disease. it is important to consider a 9am cortisol test as a random cortisol test can have low sensitivity and especially if the disease is in early stages.

Q2- Eruptive Xanthomatosis is the possible diagnosis which is due to uncontrolled diabetes and will clear on controlling the glucose levels.

other differential diagnosis would be looking for other conditions that can cause hypertriglyceridemia and diabetes such as:

Langerhans cell histiocytes

Disseminated granuloma annulare

Xanthogranuloma

dermnetnz.org. (n.d.). Eruptive xanthoma pathology | DermNet. [online] Available at:

<https://dermnetnz.org/topics/eruptive-xanthoma-pathology#:~:text=Eruptive%20xanthoma%20presents%20as%20crops> [Accessed 15 Oct. 2023].

Solak, B., Kara, R.O., Acikgoz, S.B. and Kosem, M. (2015). First and only symptom of undiagnosed diabetes mellitus: eruptive xanthoma. BMJ case reports, [online] 2015, p.bcr2015212160. doi:<https://doi.org/10.1136/bcr-2015-212160>.



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22%

341 words

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Day 5 task ▶

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